

D643

Navigating Complex Problems

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Leaders at a distribution center that recently repainted its loading docks say “shipping is bad.” Data show on-time shipping fell from 97% to 88% over two quarters; the target is 95%. The site manager has worked there 12 years. Which is the best problem statement?

- A. Shipping is bad and needs to improve.
- B. We need to hire more pickers.
- C. On-time shipping fell from 97% to 88% over two quarters, 7 points below the 95% target.
- D. On-time shipping is 9 points below target.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Patients at a dental practice that just added an espresso machine to its lobby complain about long waits. Staff add more waiting-room chairs, but waits continue. Data show appointments are booked in 15-minute slots while visits average 25 minutes. What is the root cause?

- A. The espresso machine keeps patients in the lobby.
- B. The schedule gives less time than visits actually take.
- C. Too few waiting-room chairs.
- D. Patients arrive too early.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

After a software update, a company sees several effects. Which one is a symptom rather than a root cause?

- A. The update removed the password-reset link.
- B. Training for new help-desk agents was cut.
- C. None; symptoms and causes are the same thing.
- D. Customers are calling the help desk more often.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A machine at a bottling plant that runs around the clock stopped. Why? A fuse blew. Why? The motor overloaded. Why? A bearing wasn't lubricated. Why? The lubrication pump's strainer was clogged. The plant was built in 1998. What does the 5 Whys method suggest next?

- A. Replace the equipment because the plant is old.
- B. Ask why the strainer clogged, such as whether it was ever on an inspection schedule.
- C. Replace the fuse and restart the line.
- D. Stop; the strainer is the root cause because four whys is enough.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A hospital lab team with a new director builds a fishbone diagram for mislabeled specimens. One cause: labels print with faded ink because the label printer needs a new print head. The lab recently received a state award. Which category fits this cause?

- A. Management, because the new director oversees the lab.
- B. People
- C. Method
- D. Machine (equipment)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A software company's billing system double-charged 400 customers overnight. The company recently hired a new CFO. What is the best short-term action?

- A. Rewrite the whole billing system before doing anything else.
- B. Wait for the root cause analysis before refunding anyone.
- C. Have the new CFO announce a restructuring.
- D. Pause the billing job and refund affected customers, then investigate the cause.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which action is a long-term corrective action rather than short-term containment?

- A.** Adding automated tests that block duplicate charges before code is released.
- B.** Pausing the billing job.
- C.** Posting an apology on the status page.
- D.** Refunding affected customers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A retailer with a new loyalty app logged 500 return reasons last quarter: wrong size 210, damaged 140, changed mind 60, late delivery 50, other 40. The CEO prefers bar charts. What share of returns do the top two causes account for?

- A.** 70%
- B.** 28%
- C.** 42%
- D.** 80%, per the 80/20 rule.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

In a Pareto analysis of 800 defects, the top cause accounts for 45% and the top two causes together for 75%. How many defects does the second cause account for?

- A.** 600
- B.** 240
- C.** 440
- D.** 360

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A task force at a city water utility that serves 80,000 homes is estimating repair costs. The first person to speak guesses \$2 million, and every later estimate lands near that number. The utility's logo is a blue water drop. What is at work?

- A.** Sunk-cost bias
- B.** No bias; agreement shows the estimate is accurate.
- C.** Anchoring bias
- D.** Confirmation bias

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An implementation plan at a medical-device firm with a new cafeteria lists, for the task “launch training,” two people as Accountable, three as Responsible, and five as Informed. What is the flaw?

- A.** Only one person should be Accountable for a task.
- B.** Only one person can be Responsible.
- C.** Five people Informed is too many for three Responsible.
- D.** Every task must have someone Consulted.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

In a RACI chart, a regional director must approve and answer for the outcome of a policy rollout but does none of the hands-on work. Which letter fits?

- A.** I — Informed
- B.** R — Responsible
- C.** A — Accountable
- D.** C — Consulted

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

Leaders at a distribution center ~~that recently repainted its loading docks~~ say “shipping is bad.” Data show on-time shipping fell from 97% to 88% over two quarters; the target is 95%. ~~The site manager has worked there 12 years.~~ Which is the best problem statement?

Target word: best problem statement **Decides:** Measurable gap vs. target, no solution inside.

A	Dropped element	No data or measurable gap.
B	Adjacent category	A solution, not a problem statement.
C	Correct	$95\% - 88\% = 7$ points; states the gap with data.
D	Wrong stem number	$97 - 88 = 9$ is the drop, not the gap to the 95% target.

2. Answer B Concept 2

Patients at a dental practice ~~that just added an espresso machine to its lobby~~ complain about long waits. Staff add more waiting-room chairs, but waits continue. Data show appointments are booked in 15-minute slots while visits average 25 minutes. What is the root cause?

Target word: root cause **Decides:** Scheduled time is shorter than actual visit time.

A	Camouflage	Irrelevant to appointment timing.
B	Correct	Every visit runs 10 minutes over, so delays build all day.
C	Confusion set	Chairs address the symptom; waits continued.
D	Adjacent category	Early arrivals don't explain delays past appointment times.

3. Answer D Concept 2 · transfer

After a software update, a company sees several effects. Which one is a symptom rather than a root cause?

Target word: find the symptom **Decides:** A visible effect, not the underlying condition.

A	Inverted operation	That is the underlying cause of the calls.
B	Adjacent category	A possible contributing cause, not an effect.
C	Absolute language	Separating them is the point of root cause analysis.
D	Correct	The call volume is an effect of something else.

4. Answer B Concept 3

A machine at a bottling plant ~~that runs around the clock~~ stopped. Why? A fuse blew. Why? The motor overloaded. Why? A bearing wasn't lubricated. Why? The lubrication pump's strainer was clogged. ~~The plant was built in 1998.~~ What does the 5 Whys method suggest next?

Target word: next step in 5 Whys **Decides:** The chain ended on a part, not a process.

A	Camouflage	Plant age doesn't explain the clogged strainer.
B	Correct	Pushes to a process cause that can prevent recurrence.
C	Dropped element	Fixes only the first symptom.
D	Absolute language	The count doesn't matter; the chain hasn't reached a process cause.

5. Answer D Concept 4

A hospital lab team ~~with a new director~~ builds a fishbone diagram for mislabeled specimens. One cause: labels print with faded ink because the label printer needs a new print head. ~~The lab recently received a state award.~~ Which category fits this cause?

Target word: fishbone category **Decides:** The failure is in equipment.

A	Camouflage	The director detail doesn't decide the category.
B	Confusion set	No human action is described.
C	Adjacent category	Method is the procedure; the procedure isn't at fault here.
D	Correct	A worn printer part is an equipment cause.

6. Answer D Concept 5

A software company's billing system double-charged 400 customers overnight. ~~The company recently hired a new CEO.~~ What is the best short-term action?

Target word: short-term action **Decides:** Customers are being harmed now.

A	Inverted operation	Puts the long-term fix ahead of stopping current harm.
B	Dropped element	Leaves customers harmed while analysis runs.
C	Camouflage	Doesn't address the double charges.
D	Correct	Contain the harm first, then pursue the root cause.

7. Answer A Concept 5 • transfer

Which action is a long-term corrective action rather than short-term containment?

Target word: find the long-term fix **Decides:** Prevents recurrence at the source.

A	Correct	Removes the cause going forward.
B	Confusion set	Stops the bleeding; containment.
C	Adjacent category	Communication, not correction.
D	Inverted operation	Repairs current harm; short-term.

8. Answer A Concept 6

A retailer ~~with a new loyalty app~~ logged 500 return reasons last quarter: wrong size 210, damaged 140, changed mind 60, late delivery 50, other 40. ~~The CEO prefers bar charts.~~ What share of returns do the top two causes account for?

Target word: top-two share **Decides:** $(210 + 140) / 500$.

A	Correct	$350 / 500 = 70\%$.
B	Wrong stem number	140 / 500 is the second cause alone.
C	Dropped element	210 / 500 counts only the top cause.
D	Absolute language	Pareto is a pattern, not a fixed ratio; compute from the data.

9. Answer B Concept 6 • transfer

In a Pareto analysis of 800 defects, the top cause accounts for 45% and the top two causes together for 75%. How many defects does the second cause account for?

Target word: solve for one cause's count **Decides:** $(75\% - 45\%) \times 800$.

A	Dropped element	$75\% \times 800$ is both causes together.
B	Correct	$30\% \times 800 = 240$.
C	Inverted operation	$800 - 360$ subtracts the top cause from the total instead of from the top-two count.
D	Wrong stem number	$45\% \times 800$ is the top cause.

10. Answer C Concept 7

A task force at a city water utility ~~that serves 80,000 homes~~ is estimating repair costs. The first person to speak guesses \$2 million, and every later estimate lands near that number. ~~The utility's logo is a blue water drop.~~ What is at work?

Target word: name the bias **Decides:** Estimates cluster around the first number.

A	Adjacent category	That is sticking with past spending.
B	Inverted operation	Clustering after a first guess signals anchoring, not accuracy.
C	Correct	The first figure became the reference point.
D	Confusion set	That is favoring evidence for an existing belief.

11. Answer A Concept 8

An implementation plan at a medical-device firm ~~with a new cafeteria~~ lists, for the task "launch training," two people as Accountable, three as Responsible, and five as Informed. What is the flaw?

Target word: RACI error **Decides:** More than one Accountable.

A	Correct	Shared accountability means no one truly owns the outcome.
B	Confusion set	Several people can share the work.
C	Wrong stem number	Head counts of Informed vs. Responsible don't matter.
D	Absolute language	Consulted roles are optional.

12. Answer C Concept 8 • transfer

In a RACI chart, a regional director must approve and answer for the outcome of a policy rollout but does none of the hands-on work. Which letter fits?

Target word: role from description **Decides:** Answers for the outcome.

A	Dropped element	Informed people are only updated.
B	Confusion set	Responsible people do the work.
C	Correct	The single owner who answers for the result.
D	Adjacent category	Consulted people give input but don't own the outcome.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Data-based problem statement	A good problem statement states the measurable gap between current and desired performance, without a built-in solution.	Vague complaints or statements that are really solutions.	Choosing the best problem statement.
2	Symptom vs. root cause	Symptoms are visible effects; the root cause is the underlying condition that, if fixed, stops the problem recurring.	Treating the symptom and calling it solved.	A fix was tried and the problem continued.
3	5 Whys	Keep asking why until you reach a process or system cause you can control, not just the first physical failure.	Stopping at the first failed part.	A chain of whys that ends on a component.
4	Fishbone (cause-and-effect) diagram	Causes are grouped into categories such as machine, method, materials, people, measurement, and environment.	Placing a cause in the wrong category.	A specific cause needs a category.
5	Short-term containment vs. long-term correction	Contain the immediate harm first, then fix the root cause so it doesn't recur.	Waiting on the long-term fix before stopping current harm.	An active problem hurting customers now.
6	Pareto analysis	Rank causes by frequency; a few causes usually account for most occurrences. $\text{Share} = \text{cause count} / \text{total}$.	Assuming exactly 80/20, or picking the wrong base.	Counts by category.
7	Bias in problem solving	Anchoring = over-relying on the first number heard; groups need independent estimates to avoid it.	Treating agreement as accuracy.	Later estimates cluster around the first one.
8	RACI in implementation plans	Each task has exactly one Accountable owner; Responsible do the work; Consulted give input; Informed are kept updated.	Assigning shared accountability.	A RACI chart with several A's.

Course scope (public description): Navigating Complex Problems introduces students to problem-solving techniques needed to address organizational challenges and implement solutions for desired results. Students will identify the data required to define complex problems, differentiate between short- and long-term problems, and develop strategies for resolution. Students will focus on root cause analysis, ethics and biases, communications, and the steps to strategically solve problems, while learning how to create an implementation plan for stakeholders.